

Grammaticality de-idealized

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Abstract

Why is *I have 25 years* easy to understand but ungrammatical as a way of stating one's age in English, while *Colorless green ideas sleep furiously* is grammatical despite its odd meaning? I argue that grammaticality research often runs together four questions. Does the language provide an analysis for the expression? Do the grammatical contributions of its parts fit together? Does it mark the contrasts required in this context? Is the construction established in this speech community and situation? Processing difficulty and implausibility are separate: they can lower acceptability without changing grammatical status. The Operator-Value Model of Grammaticality (OVMG) develops these distinctions and models how use, competition among alternatives, and forgetting change the constructions available to a population; a separate supplement gives the mathematical specification. The model predicts that rarely encountered constructions should respond more to exposure and framing than forms consistently displaced by an established competitor; that the absence of a form is evidence of ungrammaticality only when speakers repeatedly chose an established alternative in contexts where the missing form would have been a plausible choice; and that, as a contrast becomes moribund, the evidence sustaining it should weaken before its categorical status changes. Simulations show that specified adoption conditions can generate sharply separated population patterns; greater precision in an analyst's evidence-based estimate can't. The proposal replaces the ideal speaker's single grammaticality fact with distinct, testable claims about structure, convention, speaker experience, and change.

Keywords: grammaticality; projectibility; acceptability; preemption; entrenchment; conventionalization; usage-based grammar

AI USE. The large language models Claude, ChatGPT/Codex, Gemini, DeepSeek, and Ox Alpha served as drafting and editing aids throughout the preparation of this paper. I remain responsible for all theoretical claims, arguments, errors, and interpretive choices.

Aristotle provided formalization assistance. Model outputs used in adversarial review or formalization were checked, where applicable, by independent recomputation, local proof replay, or executable tests.

INTRODUCTION

Every competent speaker of English knows that **Can the have running* is impossible, but the source of this certainty proves remarkably elusive. What does it mean to say a sentence is ungrammatical? Consider these examples:

- (1) a. **Can the have running?*
- b. *Colorless green ideas sleep furiously.* (Chomsky, 1957)
- c. **I've finished it yesterday.*
- d. *?I saw Joan, a friend of whose was visiting.*
- e. *The bread the baker the apprentice helped made is delicious.*
- f. A: *How old are you?* B: **I have 25 years.*
- g. **Which did you buy car?*

These examples don't fail in the same way. (1a) resists a covering analysis; (1c) combines incompatible grammatical instructions; and (1f) is interpretable but lacks licensing for age predication in English. Independent relative *whose* has unsettled status, whereas centre embedding is structurally licensed but hard to process. Left-branch extraction is different again: a short, interpretable pattern that English speakers reject with striking confidence.

Different traditions emphasize different pieces: formal approaches focus on structural principles, usage-based theories on frequency and entrenchment, and processing accounts on cognitive cost. The difficulty is treating their targets as rival values on one scale. Separating them reorders the questions.

The de-idealization in the title is specific. Early generative grammar located grammaticality in the knowledge of an ideal speaker–listener in a homogeneous community (Chomsky, 1965, p. 3). That idealization bought tractability by treating variation, uncertainty, and change as external to the competence fact. Following Goodman (1955), I instead ask first what the category GRAMMATICAL lets us predict—its projectibility—and then what profile supports those predictions and what might stabilize it (Boyd, 1999; Khalidi, 2013, 2018; Millikan, 1999; Reynolds, 2026a; Slater, 2015). Naming a mechanism doesn't by itself settle either question.

The Operator-Value Model of Grammaticality (OVMG) develops that answer. It calls closed grammatical contrasts that configure what an utterance does in the conversation OPERATOR VALUES; tone, particles, word order, and inflection can all realize them (Reynolds, 2026b). The framework rests on three premises: profile, projection, and support.

1. Profile. A grammaticality claim is shorthand for a profile: a bundle of answers to several separable questions about a candidate in a situation. Is there any available analysis for it? Do its grammatically encoded contrasts agree with one another? Are the contrasts required in that situation supplied? Is the pattern a conventional resource for the relevant community, dialect, or register? And, separately, does it merely feel bad because it's implausible or hard to process? Section 2 develops these as coverage, compatibility, saturation, licensing, and subjective read-out. The examples in (1) don't instantiate one kind of “ungrammaticality”; they occupy different profiles.
2. Projection. The profile earns its standing by what it lets us predict, not by the mechanisms that hold it together: which rejections should soften under repeated exposure and which resist (1d) vs. (1g), how faithfully forms transmit, and how they change over time.

3. Support, tested separately. The dynamics module specifies candidate sufficient conditions under which usage could stabilize a population licensing state; conversational repair is a candidate *controller* – a process that detects a deviation and helps correct it – testable by its intervention signature (§3.4); and the subjective read-out of ungrammaticality is separate from grammatical status: it comprises gradient anomaly and confidence signals.

Section 1 states the problem. Sections 2–3 give the reader-facing state and dynamics theories before the diagnostic contrasts apply them in detail. Section 5 then states the predictions and positions the account against neighbouring frameworks. The accompanying supplement contains the notation, derivations, simulations, and machine-checked structural results.

1 THE IMPASSE IN GRAMMATICALITY THEORY

Three familiar problems show why a single grammaticality variable is insufficient. They concern whether an analysis exists, whether its grammatical instructions agree, and whether community convention supports it. The last divides into two questions in the model: whether the niche requires a contrast to be marked, and whether the community licenses the construction used to mark it.

1.1 THE PROBLEM OF STRUCTURAL VIABILITY

Formal production systems make membership in a set of well-formed strings categorical (Chomsky, 1957; Post, 1943). That captures cases such as (1a), which resist a viable structural analysis. It doesn't by itself distinguish them from analyzable but degraded strings. Assigning every such contrast to performance risks the circularity that Schütze (2016, p. 71) identifies: troublesome judgments can be set aside precisely because they trouble the proposed competence grammar.

1.2 THE PROBLEM OF OPERATOR COMPATIBILITY

Form and meaning can also come apart in two directions. (1b) has intact grammatical instructions despite an implausible construal, whereas the intended message in (1c) is recoverable despite a clash between the present perfect and *yesterday*. Generative semantics and Construction Grammar established that grammatical constructions make semantic contributions (Goldberg, 1995; Lakoff, 1971; McCawley, 1968); the remaining problem is to separate clashes among grammatical instructions from lexical implausibility and processing cost.

1.3 THE PROBLEM OF SITUATIONAL LICENSING

Finally, structurally viable and interpretable expressions can remain outside a community's repertoire. Sociolinguistic and usage-based work locates such patterns in community convention, entrenchment, and statistical preemption (Bybee, 2006; Goldberg, 2011; Labov, 1972). The age-predication use in (1f) is rejected in English but licensed by cognate-looking frames in French and Spanish. Structural coverage, compatibility, community convention, and felt acceptability require different diagnostics and evidence streams. The next section states those distinctions before applying them.

2 FROM AN UTTERANCE TO A GRAMMATICALITY PROFILE

The opening examples show why one acceptability scale can't do all the work. This section states the model in ordinary language; the formal supplement gives the specification, derivations, simulations, and notation. The main idea is simple: a grammaticality claim compresses several answers about a form–value pairing in a particular community and situation. The answers can come apart, and their pattern matters more than any single score.

The model separates a *STATE THEORY* from a *DYNAMICS THEORY*. The state theory asks what has to be true for a candidate to count as grammatical now. The dynamics theory asks how learning, choice, competition, repair, and forgetting could make that state persist or change. This separation blocks a common shortcut: a story about how a pattern arose doesn't by itself establish the pattern's present status, and a stable present pattern doesn't identify the mechanism that produced it.

2.1 FIX THE SITUATION BEFORE DIAGNOSING THE FORM

Every diagnosis is relative to a *CONDITIONING STATE*: the communicative situation as the participants construe it, together with the community, variety, register, or norm centre to which they orient. This state can include the activity, medium, footing, attributed speaker identity, and relevant discourse community. It isn't assumed to be obvious or externally given. Interlocutors can construe the same encounter differently, and analysts can draw the boundary too broadly or too narrowly.

For that reason, the conditioning state has to be specified independently of the judgment the model is meant to explain. A study can't label every accepting speaker as belonging to one variety and every rejecting speaker as belonging to another after seeing the responses. It needs prior evidence from community membership, usage, interactional cues, self-identification, or a registered coding protocol. If an independently motivated partition turns an apparently divided population into internally stable groups, the original conditioning state was underspecified. If no such partition does, the disagreement is a genuine population fact.

2.2 OPERATOR VALUES AND PUBLIC UPDATE

The term *VALUE* has the Saussurean sense of *valeur*: a linguistic unit is identified by what it contrasts with, what it patterns with, and which interpretations it makes available (Saussure, 1916). It doesn't mean a feature assignment or decision-theoretic value. OVMG treats grammatical knowledge as conditioned form–value relations at multiple levels of grain.

Some grammatical contrasts do more than distinguish expressions. They help determine how an utterance enters the conversational record: whether it counts as an assertion or question, who did what to whom, when it holds, what follows from it, and which referents or commitments later contributions can take up. OVMG calls these *OPERATOR VALUES* and their contribution a *PUBLIC-UPDATE ROLE*.

An operator contrast meets two conditions. Its values form a small, closed paradigm, and the paradigm has a public-update role across its ordinary occasions of use. Clause type, polarity, tense and aspect anchoring, agreement and role allocation, scope, and reference tracking are illustrative rather than exhaustive cases. The test applies to the paradigm, not to the information carried by every token. Tone, particles, word order, inflection, and suprasegmental material can realize operator contrasts (Reynolds, 2026b).

Operator membership is narrower than grammaticality. A frequent collocation or head-specific complement construction can be categorically licensed or excluded without becoming an operator. Conversely, a moribund case distinction can remain an operator while evidence for its licensing becomes less settled. Opportunity density affects that evidence, not operator membership. The operator hypothesis has its own falsifier: after closure, opportunity, and competitor strength are matched, a contrast without a public-update role should not produce the same update confusion and update-oriented repair profile as one with such a role. If it does, public-update role adds no explanatory boundary beyond constructional licensing and entrenchment.

2.3 FOUR CONSTITUTIVE QUESTIONS

OVMG treats constructions in the broad Construction Grammar sense: learned, typed associations among form, structure, content, context, and conventional constraints, at any grain from an inflectional cell to a clause pattern or an intonational contour (Fillmore et al., 1988; Goldberg, 1995; Sag, 2012). A candidate utterance can normally be assembled in more than one way from such resources. The model asks whether *some* analysis of the intended form and value passes four tests.

1. Coverage: can the language build it? There has to be at least one complete analysis of the form with the intended value. Word salad may fail here. A merely novel sentence need not: productive constructions can cover a token that has never previously occurred.
2. Compatibility: do its operator instructions fit together? The contributions that tell interlocutors how to update the conversation have to be jointly satisfiable. In **I've finished it yesterday*, the present perfect and the deictic past-time adverb provide incompatible temporal instructions. More exposure to the same analysis can't average that conflict away.
3. Saturation: are the contrasts required in this niche supplied? A community can come to treat overt marking of a distinction as obligatory. On this account, obligatoriness is the historical result of situations in which an unmarked alternative repeatedly loses to an overtly marked one. A moribund distinction can later cease to be obligatory.
4. Licensing: are the constructions used in the analysis established resources here? A form can be analyzable, interpretable, and internally compatible without being part of the relevant repertoire. English **I have 25 years* illustrates a head-and-complement convention of this kind. The intended age meaning is recoverable, but English licenses a different frame. No operator has to be mis-set for a construction to fail this test.

These tests are ordered for diagnosis but aren't four degrees on one scale. Coverage and compatibility are properties of analyses in a specified inventory. Saturation records a path-dependent community convention. Licensing is a population pattern over constructional resources. A candidate is grammatical when at least one analysis passes all four; alternative failed analyses don't contaminate a successful one. This scope matters for ambiguity, coercion, and productive novelty.

The distinction between operator value and exponent choice cuts across the four tests. **Goed* expresses the intended past value but selects an unlicensed exponent. Turkish suffix-harmony errors can have the same shape: the plural or past value remains identifiable, while the chosen allomorph isn't licensed in that phonological environment. Conversely, **depend of* is a head-specific complement convention, not an operator error. Such cases show why neither closed-class status nor categorical rejection defines the operator stratum.

2.4 POPULATION STATUS AND A SPEAKER'S EVIDENCE

The four questions concern a population profile, but no speaker or analyst sees that profile directly. Each observes a history of tokens, alternatives, omissions, repairs, and contextual cues and forms an estimate from it. OVMG keeps three things distinct: the community's current licensing pattern, an individual speaker's evidence-based estimate of that pattern, and an analyst's estimate from a possibly different sample.

This distinction adds an important second dimension to the familiar gradient mean. Two forms can receive the same middling rating for quite different reasons. One speaker may have little evidence either way; another community may be sharply divided between stable users and stable non-users. The first is epistemic uncertainty and should be sensitive to framing and new exposure. The

second is population heterogeneity and should instead appear as stable between-speaker disagreement. OVMG calls the amount and consistency of evidence behind an estimate its *EVIDENCE CONCENTRATION*: high concentration means that the estimate is well settled for or against licensing, not merely that its average is high. Mean ratings alone erase these differences.

The model also distinguishes uncertainty about the constructional inventory from uncertainty about licensing. If analysts don't know whether a productive analysis exists, that's an uncertainty about coverage. Once a viable analysis is fixed, sparse evidence about whether a community uses its nodes is a licensing question. Treating every unfamiliar token as absent from the inventory would wrongly predict that first-heard but readily productive sentences begin as ungrammatical.

2.5 WHAT THE PROFILE IS EXPECTED TO PREDICT

The profile earns its theoretical role only if observing part of it licenses expectations about cases not already used to define it. In plain terms, the analysis should tell us more than which label to attach to yesterday's rating. It should help predict how a form will be repaired, learned, transmitted, softened by exposure, or destabilized over time.

The central projection contrasts weak evidence with strong contrary evidence. A construction rare because its niche seldom occurs should remain uncertain and comparatively movable under informative exposure. A form absent despite many genuine opportunities, and repeatedly displaced by an established competitor, should produce confident and framing-resistant rejection. A processing-heavy but licensed sentence should improve with practice without a corresponding change in production or repair evidence. A moribund contrast should first lose evidential support and only later lose its categorical treatment. These targets go beyond the tests used to assign the initial profile and can fail independently.

Those failures have to change the analysis. The situation, niche, and target projection should be fixed before the outcome is inspected. If a profile then fails to predict held-out repair, exposure, transmission, or change, the projection has to be narrowed, split, or abandoned. It can't be rescued by redrawing the speech-community boundary after the responses are known.

Stabilization, maintenance, and control are distinct explanatory claims. A stable profile might be the residue of history, the result of ongoing transmission, or the effect of an active corrective process. Evidence for one doesn't establish the next. Repair remains a candidate controller until intervention shows that a deviation is registered and that the response helps restore the threatened higher-level relation.

2.6 STATUS AND THE FEELING OF STATUS

Speakers don't inspect a population state. They experience an utterance and respond to it. OVMG represents that response with two ordinary-language dimensions: how anomalous the token feels and how confident the speaker is in that reaction. Low expected licensing can contribute to anomaly, but so can processing load, lexical implausibility, garden paths, prescriptive pressure, and misidentification of the speaker or situation.

Confidence supplies information that anomaly alone misses. A preempted gap may elicit a confident "that's impossible". A winnerless cell may elicit an equally low rating but the phenomenology "I don't know how to say it". An independent-relative *whose* may feel unfamiliar while remaining movable under framing. Centre embedding can feel very bad even when the speaker accepts the construction once its parse is made clear. These aren't verbal glosses added after the fact: the model

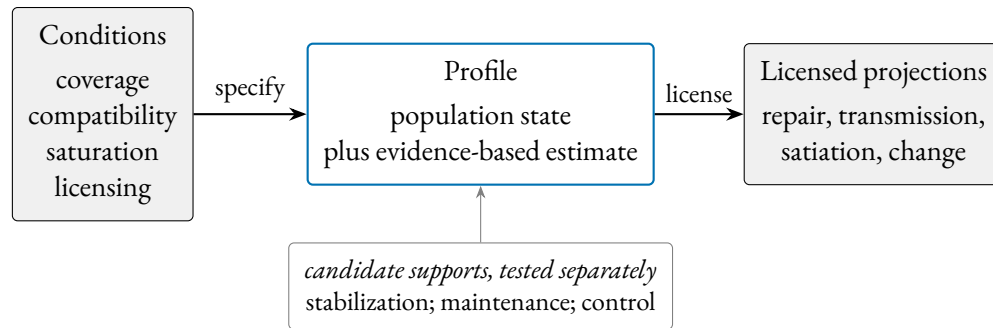


Figure 1: The projectibility-first order of explanation. The constitutive conditions specify a licensed-assembly population profile. Evidence about that profile, including how settled the evidence is and how much speakers differ, licenses the projections on the right. Stabilization, maintenance, and control are separate world-side hypotheses (below), each requiring its own evidence.

predicts different patterns of test–retest stability, confidence, exposure response, production, and repair.

A community or institution may impose a practical acceptance boundary on this graded evidence. That boundary belongs to a decision layer: changing the costs of false acceptance and false rejection should change how a hearer responds without changing the underlying licensing profile. A sharp switch in overt response would test the decision process, not prove that grammatical knowledge itself contains a numerical cutoff.

2.7 MISALIGNMENT AND THE SCOPE OF A SUCCESSFUL ANALYSIS

Hearer and speaker can disagree about which situation is in force. A form licensed in one dialect, genre, age group, or bilingual practice may be heard through another community’s expectations. The resulting anomaly can reflect a real mismatch between two stable systems rather than uncertainty within either one. Attribution matters too: a hearer who treats a token as belonging to another community, a learner error, noise, or personal overload need not respond as if it threatened the local convention.

Within a fixed situation, the existence of one successful analysis is enough. This prevents the model from rejecting ordinary ambiguity merely because one available parse fails. But the value has to be fixed independently enough to make the claim vulnerable. If analysts are free to invent a new value whenever an analysis would otherwise fail, coverage becomes vacuous. The intended value should be constrained by context, paraphrase, comprehension, and the speaker’s subsequent commitments.

2.8 MEASUREMENT REQUIRES CONVERGING CHANNELS

Status isn’t directly observed, so no single observation defines it. The empirical programme combines four channels:

- Production shows which available alternatives speakers choose in independently specified niches. A low corpus rate can reflect unavailability, low preference, or avoidance, so production alone can’t identify licensing.
- Repair shows how interlocutors respond to a putative violation. Understanding repair, update-oriented repair, overt correction, and prescriptive sanction have to be coded separately.

- Judgment and confidence measure the subjective read-out rather than status directly. Repetition, framing, and test–retest designs show how stable that read-out is.
- Situation cues support the analyst’s and hearer’s identification of the relevant community and activity. They can’t be inferred solely from the target rating.

A strong test requires at least one non-judgment indicator of licensing, an independent measure of whether changing an operator value changes the public update, and a niche definition fixed without using the target’s acceptability. The formal supplement states the corresponding observation model and its identification assumptions. The agreement corpus probe reported there illustrates the discipline: a rate per opportunity is evidence about choice, and becomes evidence about licensing only after the alternatives and avoidance option have been independently normed.

2.9 THE OPENING EXAMPLES REVISITED

The profile now gives the opening asterisks different explanations. The results are summarized in Table 1; the labels describe the decisive condition rather than mutually exclusive sentence types.

Case	Decisive diagnosis	Expected profile
<i>*Can the have running</i>	no covering analysis	confident rejection across construals
<i>Colorless green ideas sleep</i>	licensed but implausible	grammatical status with content oddness
<i>furiously</i>	content	
<i>*I’ve finished it yesterday</i>	incompatible temporal	confident rejection resistant to exposure
	instructions	of the parts
<i>*I have 25 years</i>	concentrated	recoverable meaning but confident,
	non-licensing of the	competitor-backed rejection
	English age frame	
<i>?friend of whose</i>	sparse opportunity and	low confidence and comparatively strong
	weak evidence	framing or exposure effects
<i>*Which did you buy car?</i>	putative	confident rejection only if the
	high-opportunity	opportunity and competitor
	preemption	assumptions are independently met
<i>pobedit’ first-person singular</i>	winnerless cell	low rating with ineffability rather than
		confident correction
centre embedding	implementation overload	practice can improve the experience
		without changing licensing

Table 1: Reader-facing OVMG profiles. Each diagnosis makes a different prediction beyond the judgment used to introduce it.

3 HOW LICENSING PATTERNS CHANGE

The state theory says what a present profile consists of. It doesn’t explain how the profile came to be. The dynamics module supplies candidate routes, with two constraints on the explanation. First, learning about a community isn’t the same event as changing that community. Second, absence is informative only when the missing candidate had a genuine opportunity to occur.

3.1 OPPORTUNITY, CANDIDATES, AND THE OUTSIDE OPTION

A NICHE is a recurring communicative job, such as expressing a recipient, locating an event in time, or marking an information source. Its candidate set contains the realizations speakers could plausibly choose for that job at that time. It also contains an outside option: paraphrase, clarification, silence, or avoidance. The outside option matters because failure to produce a target doesn't always mean that a competitor defeated it.

Production reflects both licensing and choice. A licensed form may be rare because speakers prefer another licensed form. An unlicensed form may be absent for the same surface reason. Conversely, a form may be scarcely attested simply because the relevant niche is scarcely encountered. The analyst has to establish the opportunity set, candidate preferences, and avoidance rate without using the target's grammaticality judgment as the coding criterion.

The evidence stream distinguishes three events. A direct error provides evidence about the target itself. An omission provides evidence only to the extent that the target would plausibly have been chosen had it been available and that the niche was correctly identified. Repair provides a further observation whose meaning depends on its type and on the relation between the error and the public update. These events enter one shared model but aren't interchangeable counts.

3.2 LEARNING AND POPULATION CHANGE

An individual updates an estimate when evidence arrives. The population changes when that evidence alters later inclusion, production, learning, or transmission. Updating an evidence-based estimate can make beliefs more precise; adoption and abandonment require a further population process. The formal dynamics specify that process, whose costs have to be motivated independently.

Depending on the adoption and retention settings, weakly supported constructions may disappear while strongly supported ones persist, producing separated population patterns. Other settings yield a single interior pattern from the same learning rule. Categorical-looking regimes depend on empirically challengeable assumptions beyond exposure.

Familiar motivations for change enter through those assumptions rather than as alternative definitions of grammaticality. Semantic reanalysis can make a new form-value relation usable, as in the development of the *going to* futurate. Social indexing can change adoption and retention, as in the retreat of the French *passé simple* from ordinary conversation. Processing economy, analogy, and iconicity can likewise alter opportunities and choices. Optimality Theory has productively represented competing pressures as ranked constraints (Prince & Smolensky, 2004); OVMG instead asks how independently motivated changes in opportunity, preference, adoption, or retention alter a community's licensing profile.

3.3 WHAT THE SIMULATIONS ESTABLISH

The simulations serve as existence and coherence checks. A deterministic approximation locates candidate long-run regimes, and a stochastic agent simulation checks whether finite populations with sampled evidence exhibit the same qualitative behaviour. Together they show that separated licensing patterns are possible when the adoption response, opportunity structure, memory, and competition are jointly strong enough. They also show that the separation disappears when those conditions are removed.

The simulations leave open the exact long-run behaviour of bounded-memory populations, how long they remain in one regime, and how networked exposure or cohort replacement affects them.

They also leave the observation channels to be connected through a single shared underlying state. The supplement reports these proof and identification obligations alongside the successful checks.

3.4 REPAIR AS A CANDIDATE MAINTAINER OR CONTROLLER

Repair can feed back into later learning, but different repairs do different work. Some secure understanding without endorsing the local form. Some replace an exponent but leave the intended operator value intact. Some correct an operator mis-setting that would otherwise change the assignment, timing, or discourse status of commitments. Overt correction can also reflect prescriptive or identity policing rather than a threat to the update.

The operator hypothesis predicts a selective response profile. Once closure, opportunity, competitor strength, and ordinary comprehensibility are controlled, mis-setting an update-configuring contrast should disproportionately elicit update-oriented repair. A categorically unlicensed but non-operator construction may instead be corrected as a conventional form choice. The prediction concerns the repair target. Non-operators can still be categorically policed.

Calling repair a *controller* requires more than observing that correction and stability coexist. An intervention would have to show that the system registers the relevant deviation, changes lower-level behaviour in response, and thereby restores the threatened population relation. Evidence that removing repair weakens transmission would establish maintenance; evidence of the full deviation-sensitive loop would establish control.

3.5 FOUR PREDICTED TRAJECTORIES

The dynamics yield four qualitatively different trajectories that would be collapsed by a single frequency measure.

OBLIGATORIFICATION. An overt contrast becomes obligatory when unmarked realizations repeatedly lose in well-attested niches and that pattern persists long enough to become a community convention. English progressive marking and Turkish evidential marking are proposed cases. The claim is historical and niche-specific: the model can't infer obligatoriness from the analyst's present uncertainty or from the absence of data. Once established, obligatoriness also doesn't vanish with the first contrary token; recovery has to persist before the convention changes. The proposal would be weakened by a contrast that became obligatory despite a well-attested niche in which unmarked alternatives hadn't repeatedly lost, or that ceased to be obligatory without a prior loss of evidential support.

WINNERLESS CELLS. A dense niche can lack a conventional winner. Speakers may paraphrase or avoid the job. Because avoidance isn't readily attributable to any one candidate, it doesn't by itself lower support for a particular candidate; it mainly starves all of them of evidence. Low licensing with low confidence also requires independently motivated weak starting support or diffuse negative evidence. The resulting profile is ineffability rather than a competitor-backed ban. Russian defective paradigms supply the clean model; English **amn't* is likely mixed because a competitor and avoidance both matter.

DESTABILIZATION. A moribund contrast can lose evidential concentration before its average status changes much. If a shared evidence stream disappears, individual histories can drift back toward starting expectations. Between-speaker divergence follows early only when a further source of heterogeneity (different networks, cohorts, starting expectations, or exposure histories) is present. Concentration loss is the direct prediction; population dispersion is conditional.

ACTUATION AND APPARENT CATEGORICAL EXCLUSIONS. Small changes in opportunity, competitor strength, analogy, repair, or social ascription can move a construction across a boundary between long-run regimes. These inputs supply candidate actuation routes; identifying a historical trigger requires independent diachronic evidence. At the other extreme, persistent preemption in a dense niche can generate a highly concentrated rejection even when a compatible analysis remains available. Apparent categorical exclusions don't always require an additional structural ban; they require independent evidence that the niche was rich, the target was a live candidate, and a competitor repeatedly won.

The dynamics succeed only if independently measured opportunities, alternatives, repair types, evidence histories, and adoption costs predict the distinct trajectories. The formal supplement supplies the quantitative claims and reproducible simulation contract.

4 DIAGNOSTIC CONTRASTS

The model's questions now let us compare cases without forcing them onto one severity scale. The labels below name common routes through the architecture, not additional primitives. Each case also supplies evidence or a falsifier that the short diagnoses in Table 1 leave implicit.

4.1 CONDITIONING AND COMMUNITY CONVENTION

Form–value relations occur at every level of grammatical organization, from morphemes to abstract clause patterns (Goldberg, 1995). Individual forms can participate in several such relations: English past-tense morphology usually locates an event in the past, but it can also mark social distance (*Could you help me?*) or low likelihood (*If I went tomorrow...*) (Kilgariff, 2004). A diagnosis concerns a relation in a specified situation, not a form with one context-free meaning.

Communicative situations overlap and shift with activity, audience, durable social position, and the norm centre participants orient to (Wiese, 2023, p. 3). Multiple modals such as *I might could help you with that* are productive for some American English speakers and rejected by others (Morin & Grieve, 2024). Spanish–English bilingual practice likewise licenses some recurrent mixtures while excluding others (Toribio, 2001). These sources of variation still have to be fixed independently by the procedure in §2.1.

Convergent acceptance is evidence about a licensed repertoire. Status has to be specified independently enough to identify contested innovations, stable dialect differences despite mutual intelligibility, and performance errors that pass unnoticed. The form, value, situation, and comparison class can't be inferred from the response alone.

4.2 SATURATION OF OBLIGATORY CONTRASTS

Languages differ in which distinctions their constructions require speakers to mark. Standard English normally requires progressive aspect for an event presented as ongoing (*She is studying right now*). Standard French can describe the same situation with the simple present (*Elle étudie maintenant*) because progressive marking isn't obligatory there. Evidential distinctions provide a parallel cross-linguistic contrast: some languages grammaticalize an obligatory evidential paradigm, whereas English usually leaves source marking to optional lexical or modal resources. OVMG treats these obligations as path-dependent conventions whose establishment and possible loss are addressed by the obligatorification trajectory in §3.5.

4.3 OPERATOR-VALUE COMPATIBILITY

Compatibility asks whether the operator contributions of a proposed analysis can jointly hold. The individual pieces may be familiar and the intended message recoverable even when their grammatical instructions clash.

Consider the temporal incompatibility in:

- (2) * *I've finished it yesterday.*

Here, the present perfect construction contributes a reference interval anchored to present relevance, while *yesterday* contributes a deictically past interval. Unlike lexically incongruous combinations like *colorless green ideas*, which remain grammatical because the clash doesn't mis-set an operator value, this example shows failed unification between grammatically encoded temporal contributions.

Sometimes compatibility failures involve conflicting information-packaging requirements:

- (3) * *Who did the lifeguard who saved _ work in New Jersey?* (Cuneo & Goldberg, 2023)

The clash is in grammatically encoded information packaging: the same participant is simultaneously focused through fronting *who* while being backgrounded by the relative clause construction (Cuneo & Goldberg, 2023).

These examples also delimit compatibility. The compatibility condition treats operator-value conflict as failed unification among partial operator assignments: tense–aspect anchoring where tense–aspect is obligatory, argument-linking requirements where a construction licenses roles, information-packaging instructions where a construction grammatically encodes focus or backgrounding, and indexical contrasts only when they've entered a closed public-update repertoire. This makes compatibility partly derivative of licensing facts at the feature level, keeping arbitrary lexical implausibility out of grammatical incompatibility.

4.4 PROCESSING COST WITHOUT STATUS FAILURE

Processing demands can trigger feelings of ungrammaticality even when a construction passes all four status tests. Centre embedding provides the clearest contrast:

- (4) *The bread the baker the apprentice helped made is delicious.*

Dependency locality is a well-documented source of this cost: integration grows harder as dependencies span more intervening material, especially when new referents intervene (Gibson, 2000, 2026; Grodner & Gibson, 2005). Specialized language networks and domain-general systems can both contribute to the resulting experience (Fedorenko et al., 2024). Avoidance can later affect transmission, but that later route into change is separate from present status. A highly licensed construction can remain grammatical while its anomaly signal stays high.

Socio-pragmatic indexicality has a parallel but distinct role. Forms such as Río de la Plata *vos* can identify region, relationship, or stance as well as the addressee (Bertolotti, 2016; Eckert, 2012; Silverstein, 1976). Phonological cues likewise support rich identity expectations without thereby triggering ungrammaticality (Babel, 2025). Such information helps participants identify the relevant situation and norm centre. It becomes an operator contrast only when it also belongs to a closed, public-update paradigm.

4.5 LICENSING, OPPORTUNITY, AND PREEMPTION

Even a well-formed construction with a clear meaning can fail because the community doesn't license it in the relevant situation.

The age-predication use of **I have 25 years* is the opening case. English licenses *have+years* for duration, remaining time, or experience, but ordinary age predication uses the *be* frame. The intended age is recoverable and no operator value is mis-set; the decisive failure is constructional licensing in a niche where an established competitor is encountered constantly.

- (5) * *We sheared three sheeps.*

The regular plural marking has a compositionally recoverable meaning and is structurally parallel to other English plurals, but the irregular form *sheep* is entrenched across English-speaking communicative situations, making the regularized version unacceptable. Without a metaphorical or playful justification (as in *the black sheeps of the family*, where the irregular plural might signal a figurative usage), the utterance is ungrammatical.

Situational licensing operates independently from the other conditions. A construction may have complete operator compatibility and minimal processing demands but still be excluded because a different form has been conventionalized for that meaning. Extreme rarity makes the distinction clear. Some constructions are so infrequent that speakers lack a shared consensus about their status.

Independent relative *whose* shows why rarity isn't automatically negative evidence. Its component resources are familiar: independent genitives are possible (*Mine was visiting*), independent interrogative *whose* is grammatical (*Whose was open?*), and dependent relative *whose* is common (*the student whose friend was visiting*) (Hankamer & Postal, 1973). Yet their combination is so unusual that Hankamer and Postal (1973) deem it non-existent:

- (6) ? *I saw Joan, a friend of whose was visiting.*

(adapted from Huddleston & Pullum 2002, p. 472)

The construction requires an accessible possessor, a predictable possessum, an appropriate information structure, and an ellipsis-licensing environment at once (Reynolds, 2024). Those conditions rarely converge. The COCA check reported here finds no token (Davies, 2024), but raw component frequency is the wrong denominator: separate occurrences of independent and relative *whose* don't create opportunities for the combined niche. The absence supplies little negative evidence. Speakers should remain uncertain and comparatively responsive to framing or exposure, although high surprisal can add processing difficulty even for speakers who recover the analysis (Shain et al., 2020).

This contrasts with putative preempted cases like left-branch extraction, discussed next, where the relevant niche appears often and the construction is systematically untaken.

APPARENT CATEGORICAL EXCLUSIONS. Some constructions face rejection so persistent that it *appears* categorical. Left-branch extraction is the textbook case:

- (7) * *Which did you buy [___ car]?*

The intended meaning is easily grasped; nothing in the semantics or pragmatics prevents understanding the speaker's intent. Nor does the construction seem excessively complex to process. Yet

English speakers categorically reject it, and Snyder (2022, p. 22) treats this kind of pattern as a core case of “ungrammaticality”.

Under OVMG, such patterns are candidate preempted gaps: situational licensing is driven toward zero by persistent preemption when the communicative job is common, an established competitor wins, and the counterfactual choice probability of the missing form is high. For English left-branch extraction, the natural competitor is *Which car did you buy?*. The corpus absence is suggestive but not decisive; the missing ingredient is independent evidence about how often speakers would choose the discontinuous form if it were available under permissive framing. Additional processing penalties, such as systematic garden-path reanalysis when the bare interrogative phrase is encountered, can strengthen the subjective categoricity without any independent structural constraint. The relevant adoption dynamics are summarized in §3.5; the supplement gives their formal specification.

Cross-linguistic comparison has to be made over comparable niches, not over the descriptive label LEFT BRANCH (Haspelmath, 2010). Discontinuous nominals should be easier to license where inflection makes the dependency recoverable, a productive discontinuous pattern supplies scaffolding, and a discourse niche makes separation useful. Whether the relevant constructions in case-rich, article-less languages are comparable to English remains open (Reynolds, 2026c).

Several diagnostic criteria help identify these preempted-gap constructions:

1. Persistent non-licensing: Independent production and repair evidence continue to support non-licensing across occasions.
2. Categorical rejection: Speakers find the constructions simply impossible, with high confidence rather than mere uncertainty.
3. Resistance to satiation: Unlike processing-heavy cases, repeated exposure doesn’t improve acceptability.

Prediction: If LBE is a high-concentration preempted gap, framing tokens as intentional dialectal or ingroup resources should do little to move judgments; the same manipulation should matter more for low-concentration cases such as independent relative *whose* or defective cells. A substantial framing effect for LBE, or independent evidence that speakers would rarely choose the discontinuous form even if it were available, would count against the preempted-gap diagnosis.

Expressions that lack a complete analysis instead fail structural coverage. The preempted-gap diagnosis is restricted to expressions such as LBE for which a viable analysis is independently supported and the explanatory work can fall to the dynamics of situational licensing.

WINNERLESS CELLS: PARADIGM DEFECTIVENESS. Preempted gaps have a winning competitor. Defective paradigm cells don’t. They leave a high-opportunity cell empty without any single form winning it. A clearer case is the Russian first-person singular non-past of *pobedit* ‘win’: speakers avoid the expected cell and resort to periphrasis instead. The English *amn’t* gap is diagnostically useful but less clear, because broader niches include *I’m not* as a competitor, while narrower contracted-negator niches make the cell look winnerless (Broadbent, 2009; Pertsova, 2016; Pertsova & Kuznetsova, 2015; Sims, 2023; Thoms et al., 2025).

In the Russian case, the niche is common and candidate forms are readily identifiable, but no candidate accumulates enough positive evidence to settle as the form. The source can be structural uncertainty, lexical restriction, learning dynamics, or avoidance, with the mixture varying across cases (Sims, 2023). In this account, defectiveness is a third licensing profile: high opportunity with evidence divided among candidate forms, leaving no candidate stably established or confidently excluded.

The phenomenology matches: speakers report not knowing how to say it (ineffability), not the confident “that’s wrong” of a preempted gap. The dynamics in §3.5 explain why avoidance needn’t provide confident negative evidence against any particular candidate. Defective cells should remain more movable under framing than preempted gaps.

The contrasts also explain why an intermediate rating has no unique diagnosis. It can reflect weak licensing evidence, a strong anomaly response to a licensed form, or disagreement about the conditioning state. Lexical-semantic oddity without an operator clash, as in *colorless green ideas*, likewise belongs to the subjective read-out rather than grammatical status. The profile table in §2.9 summarizes the resulting routes and their different projections.

4.6 THE SUBJECTIVE EXPERIENCE OF GRAMMATICALITY

Speakers register putative violations through a read-out that can diverge from conventional status.¹ The anomaly channel resembles other metacognitive feelings such as the FEELING OF KNOWING (Hart, 1965); the confidence channel records how settled the speaker takes the reaction to be. The distinction also recalls Sapir’s “form-feeling” and is consistent with self-monitoring in Broca’s aphasia (Oomen et al., 2005; Sapir, 1921, 1927). This response can help speakers detect a violation, but it doesn’t define grammaticality (Fanselow, 2021).

Status has to be inferred from converging measures, much as temperature is inferred from the behaviour of instruments. Ratings and factor analyses primarily reveal the subjective read-out; they bear on grammar only when they converge with independent indicators of licensing, compatibility, and confidence (§2.8). Raw language-model probability has the same limitation, and even minimal-pair contrasts are informative only when the intended message is controlled (Hu et al., 2026).²

Misattribution makes the separation visible. A grammatical construction can feel wrong after a garden path:

- (8) *The old man the boats.* (Ritchie & Thompson, 1984)

The initial noun-phrase analysis of *old man* produces nonsense; on the licensed analysis, *the old* is the subject and *man* the verb. Conversely, a clear intended meaning can mask an unlicensed analysis:

- (9) * *In Michigan and Minnesota, more people found Mr. Bush’s ads negative than they did Mr. Kerry’s.* (Pullum, 2009)

The first clause invites a comparison between numbers of people, but the pronoun *they* in the second supplies the wrong comparison structure. Hearers often recover the intended message without registering that failure.

5 THEORETICAL IMPLICATIONS

The architecture earns its keep through contrasts that simpler descriptions don’t predict. This section begins with those tests, then asks how the proposal relates to macro-typology and neighbouring theories. Licensing, feeling, and their possible supports retain separate evidence streams throughout.

¹The distinction has a philosophical analogue in Cavell’s (1969) distinction between knowledge and acknowledgment. For LLM-based research it blocks an easy inference: judgments that correlate with human ones don’t establish that a model acknowledges the norms rather than processing correlated patterns (Techio, 2026).

²Nefdt and Ladyman (2026) distinguish proposals that locate linguistic patterns in outputs, cognition, model-internal compression, or a plurality of models. OVMG’s narrower claim is that grammaticality, for the purposes tracked here, is a projectible pattern in situation-indexed licensing profiles.

5.1 PREDICTIONS

The model's distinctive predictions come from separating positive evidence, opportunity structure, how likely a form would have been chosen had it been licensed, how settled the licensing evidence is, and repair flow. They concern concentration contrasts, zero-evidence learning, L2 transfer, moribundity, repair type, and change by re-licensing.

SATIATION FRAMING, AS A RANK ORDER. Framing should matter most where the evidence is least concentrated. Once the relevant niche has been identified independently, confirmed preempted gaps and **sheeps* should show the least movement; entrenched but uncommon constructions should show more; and independent relative *whose* and clean winnerless cells such as *pobedit'* 1SG should show the most. Participants told that tokens of a low-concentration form come from an intentional dialectal, literary, or ingroup resource should show more improvement than participants told the same tokens are errors; the same manipulation should leave a confirmed preempted gap unmoved. Lu et al.'s (2024) meta-analysis of syntactic satiation in extraction from islands supplies a nearby benchmark for extraction phenomena, not a direct measurement of left-branch extraction. The dependent variables are rating change, evidence-confidence change, decision-confidence change, later repair behaviour, and elicited production. If, under matched opportunities, low-concentration forms move less than confirmed preempted gaps, the predicted rank order reverses and the concentration account fails.

ARTIFICIAL-LANGUAGE LEARNING. An artificial-language-learning experiment can hold positive evidence constant at zero while manipulating opportunity-set size. Learners could see a system in which a target candidate is never produced. In the high-opportunity condition, competitor candidates repeatedly occupy the relevant niche and the target would otherwise be a plausible choice; in the low-opportunity condition, the niche rarely arises or the target would seldom be chosen even if available. OVMG predicts categorical rejection in the first condition and uncertainty in the second, even though both conditions contain identical zero positive evidence. The design operationalizes the supplement's sparse-opportunity versus dense-preemption contrast and is compatible with artificial-language learning paradigms that manipulate simplicity and communicative structure (Culbertson & Kirby, 2016).

L2 PREEMPTION TRANSFER. Early L2 rejections should track the accumulated weight of counterfactually informative omissions in the L1, not just L2 frequency. A learner whose L1 strongly preempts a candidate in a high-opportunity niche should reject the L2 analogue early, even if the L2 input frequency is low enough to leave native speakers uncertain. Conversely, low L1 preemption should make learners more willing to treat rare L2 patterns as unresolved rather than impossible. As learners join the L2 speech community, their situation ascriptions and constructional hierarchy should shift toward the target norm-centre. This view aligns with Selinker's (1972) concept of INTERLANGUAGE, but gives a more specific prediction: transfer is strongest where the L1 has accumulated many attributable, counterfactually informative omissions. If early rejections instead track L2 input frequency once L1 opportunity and competitor history are matched, the transfer claim fails.

CONCENTRATION LOSS BEFORE CATEGORICAL DECLINE. For a contrast going moribund, the direct destabilization result (§3.5) is loss of excess evidence concentration as opportunities thin. Longitudinal designs should measure confidence, test–retest stability, and sensitivity to controlled exposure alongside means and production. A reliable drop in those concentration-sensitive measures while the population rate remains approximately stable would support the proposed precursor. Inter-speaker judgment dispersion is a separate, conditional target. When speakers begin with different expect-

ations but share the same thinning evidence, the spread between them grows more slowly than the average falls. A dispersion-leading result should be predicted only when cohort replacement, unequal network opportunity loss, heterogeneous retention, or a specified judgment mapping supplies faster divergence.

SOFT ALTERNATIONS AS A NON-BIMODAL CONTROL. The conditional dynamics (§3.3) predict separated licensing only where the observation process and independently motivated adoption costs support it; a both-licensed alternation should stay unimodal, its variation carried entirely by selection among two licensed variants and any outside option. The dative is the worked case (§5.5): a corpus production model is well calibrated on the recipient choice within its high-frequency core, the graded, single-peaked signature OVMG predicts when both variants are securely licensed. The alternation is non-categorical through optionality, not the uncertainty of a winnerless cell (§3.5): the niche is common and both variants are licensed, so judgments should be confident and gradient rather than ineffable.

REPAIR TYPE. The repair streams (§3.4) predict which part of an utterance is targeted. Update-oriented repair should increase with niche opportunity and with the extent to which changing the contrast changes the public update, conditional on the other factors. Form correction can remain strong for a categorically excluded construction or exponent even when the intended public update is preserved. Opportunity, update consequences, and licensing concentration are independently estimable from opportunity-annotated corpora, comprehension probes fixed before the repair data is inspected, and the non-rating licensing streams. If matched non-configuring contrasts draw the same update-oriented repair as operator contrasts, or if that repair tracks social indexing rather than measured update consequences, the operator-directed controller claim fails.

CHANGE BY RE-LICENSING. Hard unification makes a commitment that a weighted-penalty model doesn't: no amount of entrenchment can make the same clashing operator assignment grammatical. Apparent change in cases like **I've finished it yesterday* should come either from read-out effects or from re-licensing. Where present-perfect combinations with definite past-time adverbials are attested in a community (Werner, 2013a, 2013b), change has to proceed by *re-licensing*: a perfect construction with a different operator contribution enters the repertoire, potentially producing judgment distributions that are bimodal across communities and categorical within speakers, not gradual within-speaker reduction in the status cost of the original clash. A soft-constraint model predicts the opposite signature. Judgment-distribution data from the relevant contact varieties decides it.

This contrasts with preempted gaps. A preempted gap can re-open when changed preferences, opportunities, or target evidence move the population across a transition boundary: average licensing then rises through the middle, producing within-speaker gradient during the transition and potentially a community S-curve.

Because production supplies later evidence and retained evidence affects later adoption, some settings permit a stronger, history-dependent pattern. Two communities under the same current external conditions can remain stably different because their inherited evidence states place them on different trajectories. Licensing can then disappear at one point as conditions worsen but return only after conditions improve beyond a different point. This history dependence predicts distinct loss and recovery boundaries.

The prediction is conditional and requires longitudinal evidence. Effective exposure and the evidence speakers require before adopting or retaining the form have to be estimated without using the

transition points; if matched loss and recovery trajectories retrace the same boundary, a steep but reversible response is the better account.

Compatibility failures can't re-open that way; they require a distinct construction with a different operator contribution. The identification discipline is the same as elsewhere: the re-licensed construction has to be diagnosed before the judgment data is used, for example by showing that it supports other definite past-time adverbials, scope interactions, or ellipsis patterns predicted by the new operator contribution.

INTONATIONAL OPERATORS. Nothing in the model privileges segmental material, so intonational patterns are predicted to count as operator exponents wherever they form closed, update-configuring repertoires. Opportunity density affects the concentration of their licensing evidence, not membership. The English polar-question rise and nuclear-accent placement configure what an utterance does to the common ground much as tense and agreement do (Reynolds, 2026b). The Turkish suffix-harmony analysis in Appendix A gives the parallel case for segmental exponent choice: phonological material matters for grammaticality when it realizes an operator exponent. The corresponding intonational test compares repair targets. After opportunity and competitor strength are matched, mis-set intonational operator values should produce more update confusion and update-oriented repair than value-intact exponent errors or non-operator accent shifts. Categorical correction and resistance to framing can occur in any of those cases, so neither response diagnoses operator membership by itself.

5.2 MACRO-TYPOLOGICAL CONSTRAINTS ON GRAMMATICAL DESIGN

Macro-typological work shows both recurrent constraints and substantial community-specific freedom. Verkerk et al. (2025) test 191 implicational universals against Grambank's 2,430-language sample while controlling for genealogical and areal non-independence. Only about a third remain statistically supported; the survivors concentrate in hierarchical and relatively narrow word-order patterns, often in configurations that languages are more likely to evolve into than out of.

OVMG treats such results as candidate attractors in the design space, not as proof that a comparative feature is a language-internal grammatical object. That stronger claim requires a comparandum-to-realization mapping: the feature, its realization in each language, and the held-out language or diachronic outcome to be predicted have to be specified (Reynolds, 2025). Recurrent configurations may reflect shared cognitive and communicative pressures, while the many unsupported universals leave room for contingent community conventions. Comparative evidence locates candidate stable regions; OVMG supplies conditional routes by which local licensing could reach them.

5.3 RELATIONSHIP TO GENERATIVE GRAMMAR

Generative work made a central contribution by showing that grammaticality can't be reduced to semantic plausibility, processing ease, or frequency of attestation. Chomsky's *Colorless green ideas sleep furiously* made the contrast visible: speakers can recognize syntactically well-formed sentences even when they're semantically bizarre. Generative analyses explain apparent categorical exclusions like the determiner-head discontinuity in English left-branch extraction (**Which did you buy car?*), and they capture the fact that such sequences remain unacceptable even when their intended meaning is clear and processing demands are low.

OVMG retains the lesson. Judgments reflect systematic patterns (Dennett, 1991); those patterns can't be reduced to meaning or processing alone; and certain syntactic configurations appear to be excluded regardless of context. The earlier diagnostics for center embeddings and preempted gaps

build on ideas about recursive structure and acknowledge the generative discovery that some exclusions are systematic. They relocate the explanation in coverage, hard compatibility, saturation, and concentrated exclusion.

The same point governs the account's use of formal evidence. A generative analysis can supply a real warrant when it specifies the target, diagnostics, evidence, and failure conditions for a grammatical contrast. OVMG has a distinct target: the community-level licensing profile that makes some form–value relations categorically licensed or excluded. An I-LANGUAGE analysis can still explain the structural contrasts that enter that profile.

Independent relative *whose* marks the difference in explanatory target. Generative work documents why its components appear syntactically available; OVMG adds the low-confidence, framing-sensitive prediction of a rare niche. Age predication supplies the parallel lexical-constructional case: structural possibility doesn't determine which form–value associations a community establishes (§4).

5.4 RELATIONSHIP TO CONSTRUCTION GRAMMAR

Construction Grammar (CxG) treats learned form–meaning pairings at every grain as grammatical objects (Fillmore, 1988; Fillmore et al., 1988; Goldberg, 1995, 2019; Kay & Fillmore, 1999; Sag, 2012). OVMG inherits that ontology and the associated rejection of a clean core/periphery split. Its additional distinctions concern what a construction does in the present profile: whether the community licenses it, whether it belongs to a closed public-update paradigm, whether the intended operator value was selected, and whether that value used a licensed exponent.

Constructional licensing is broader than operator membership. A community can categorically exclude age-predication *have* or a head-specific complement without any operator value being wrong. Conversely, phonological, gestural, lexical, morphological, and syntactic material can all realize an operator contrast when they enter a closed public-update repertoire (Reynolds, 2026b). Opportunity and competitor evidence predict the concentration of licensing; public-update divergence predicts the narrower comprehension and repair profile.

5.5 RELATIONSHIP TO USAGE-BASED APPROACHES

OVMG shares with usage-based approaches (UBA) (Bybee, 2006, 2007, 2010) the idea that linguistic knowledge develops from patterns of actual language use rather than from an autonomous formal system. Both perspectives reject the notion that grammaticality can be reduced to abstract rules operating independently of meaning and context.

Several pieces of the model come directly from this tradition. The speaker/population split restates Schmid's (2020) separation of ENTRENCHMENT, the cognitive reorganization of a speaker's knowledge, from CONVENTIONALIZATION, the social establishment of a community's regularities. A speaker's own repertoire, the population licensing rate, and an analyst's estimate of that rate are separate objects. The proposed population change process is motivated by Croft's (2000) utterance-selection theory, in which variants are replicated and selected within a community; characteristic trajectories of change are already worked out in that lineage (Blythe & Croft, 2012).

Frequency per opportunity is central to Stefanowitsch's (2006) case that corpora contain negative evidence, where a form counts as “significantly absent” rather than “accidentally absent” only once its observed frequency is weighed against the frequency expected given its opportunities. Variationist sociolinguistics has normalized by opportunity since Labov's principle of accountability (Labov, 1972).

OVMG adds their joint formalization and a prediction. It separates a candidate’s availability from a speaker’s choice among available candidates. That division puts a quantity on the absence that Stefanowitsch left open: he showed how to establish that a form is significantly absent but stressed that significant absence “does not, in itself, provide any clues as to why”. A significantly absent form may be unlicensed, licensed but dispreferred, or avoided together with its competitors, and those diagnoses project differently. The formal supplement states the factorization and the restricted conditions under which an attributable omission provides approximately direct evidence of non-licensing.

Speaker-level evidence estimates feed the adoption–retention process described in §3.3. The relation to Goldberg’s (2019) COVERAGE and statistical preemption is one of formalization in the same spirit. Goldberg’s “coverage” is overall frequency support; structural coverage here asks whether a complete analysis exists. Her “explain me this” puzzle (a high-coverage double-object frame preempted by the prepositional dative) is a case in which accumulated, attributable competitor wins exclude the target in a high-opportunity niche.

The dative alternation gives the licensing-versus-selection factorization its cleanest empirical case, because both variants are licensed and the whole contrast lives in selection. Bresnan et al. (2007) model the choice between *give Kim the book* and *give the book to Kim*; a frozen version of such a model can be rescored on later British conversation in the Spoken BNC2014 to estimate selection directly. Because both realizations are grammatical, nothing in the corpus probability bears on grammatical status: it estimates the probability of an NP recipient conditional on a token that has already entered the annotated alternation sample, a conditional production probability rather than a probability of grammaticality. The released rows fix realization inside that annotated frame, the innermost of a set of nested opportunity sets; the broader licensing set is a different denominator, and reaching it marks a new target rather than a larger sample of the same one.

The conditioning that transports is projectible in Goodman’s sense (Goodman, 1955): length, animacy, definiteness, and pronominality carry across varieties, while verb base rates travel worse, and even the transportable relations become overconfident out of domain. A relation can be projectible and still project poorly under a population shift, because selection is variety-relative.

This both-licensed control also exposes the identification limit: from inside an annotated alternation sample, an unlicensed gap and a merely dispreferred form can look alike. Telling them apart requires evidence of another kind, not more tokens from the same sampling design.

One caution follows from Ambridge et al. (2015), who find that overall form frequency (entrenchment) predicts the retreat from overgeneralization more robustly than competitor frequency (preemption), and who propose collapsing the two into a single competition process. OVMG keeps them analytically separate as direct error observations and counterfactual choice observations. They have to be analysed jointly rather than added as interchangeable failure counts (§3.1). Whether one underlying competition process accounts for both remains an empirical question rather than a stipulation.

Independent relative *whose* supplies the complementary low-opportunity case (§4). OVMG thus keeps the usage-based role for frequency and entrenchment while predicting when rarity is informative preemption and when it remains uncertainty.

5.6 RELATIONSHIP TO LOGICALITY OF LANGUAGE ACCOUNTS

Logicity-of-language accounts explain some restrictions by positing a system that identifies meanings made trivial by logical or closed-class vocabulary (Chierchia, 2013; Fox, 2000; Gajewski, 2009).

Their central challenge is that ordinary tautologies and contradictions can be acceptable. To address this, Del Pinal (2019) allows context-sensitive rescaling of open-class terms: the second occurrence of *raining* in *It's raining and it isn't raining*, for example, can be understood as 'raining hard'.

This account targets restrictions tied to logical vocabulary. OVMG also has to distinguish absent analyses (1a), conventional exclusions (5), and rare, weakly evidenced constructions (1d). It treats interpretive rescaling as one possible route to a compatible analysis, while keeping that analysis separate from community licensing and subjective processing effects (§2.6). The two approaches overlap on some compatibility cases but make different commitments about the source and scope of grammatical exclusion.

5.7 RELATIONSHIP TO RELEVANCE-THEORETIC ACCOUNTS

Scott-Phillips (2026) argues that acceptability intuitions arise as by-products of systems for interpreting communicative acts, with unacceptability reflecting an inability to interpret an utterance consistently with presumptions of communicative efficiency. OVMG shares the view that communicative practice shapes linguistic conventions, but treats the resulting form–value relations as constraints that needn't reduce to present efficiency.

The empirical difference concerns what has to be established independently. A relevance-theoretic account has to specify why a case is inherently impossible to interpret rather than merely difficult. OVMG has to identify the relevant construction, situation, and any operator clash before using the judgment as evidence. Novel constructions becoming acceptable or unacceptable provide the clearest comparison: an efficiency account predicts movement with changing interpretability, whereas OVMG allows interpretability to remain high while community licensing or operator compatibility changes.

6 LIMITATIONS AND FUTURE DIRECTIONS

The account has five main limits. The first is its reliance on the concept of “communicative situation”. Speakers move among overlapping contexts that shift with the immediate setting, durable social positions, and the norms they orient to. The account builds in this fluidity, but operationalizing these dimensions, and measuring their influence on grammatical stability, will take more precise methods.

A further limit lies in the formal model. The supplement uses an approximate evidence filter, declares rather than estimates the adoption criterion, and approximates the expected population transition. Its long-run results for bounded-memory populations remain unproven. Networked exposure, cohort replacement, and a shared observation model for production, repair, ratings, and confidence remain future work. The emergence claim is conditional on those interfaces rather than a generic consequence of learning.

Stylistic variation and individual preference mark another boundary. Most choices that fall within grammatical acceptability aren't cases of (un)grammaticality at all. They matter for this account only when they become tied to communicative situations strongly enough to affect licensing, repair, or categorical rejection. The account still has to explain how stylistic preferences become grammatical licensing facts.

Constructional coverage supplies a further limit. The paper argues that many apparently arbitrary restrictions are either compatibility failures, saturation failures, or concentrated non-licensing under historical preemption. Cases that resist those treatments should be analyzed as failures to build

a complete covering analysis, or else as evidence that the constructional inventory has been specified too narrowly.

The account also carries a memory burden. It asks speakers to retain situation-indexed information about forms, values, constructional neighbours, opportunities, competitors, and repair histories. Compositional inheritance means that first-heard tokens needn't begin at zero, and pooling across related constructions could reduce the burden further, but neither compression process is yet estimated. How much speakers retain, and how far inheritance or pooling compresses it, remains an empirical question.

6.1 METHODOLOGICAL CONSIDERATIONS

Experimental syntax supplies tools for separating the proposed evidence streams (Schütze, 2016). Corpus rates have to be normalized by independently specified opportunities; judgments should be accompanied by confidence, test–retest, and exposure measures; and processing measures can test whether practice lowers implementation cost while production and repair remain stable. Sociolinguistic fieldwork fixes the relevant community and norm centre, while longitudinal and historical data test whether evidence concentration changes before categorical treatment does. No one channel identifies licensing alone.

A DECISION-LAYER DISCONTINUITY CHECK. A community or task can place a practical boundary on a graded estimate of licensing. A sudden change in response type near an independently estimated boundary would test whether asymmetric losses shift hearers from acceptance to repair, clarification, or sanction beyond the smooth effect of their own estimate. The general design is a regression-discontinuity problem (Calonico et al., 2014; Lee & Lemieux, 2010); multinomial and covariate-adjusted versions are the needed extension (Lazzaro, 2026). In a confirmatory test, the boundary has to be estimated and registered before the outcome data is inspected. A smooth response would weaken the decision-layer switch, not the licensing-state theory itself.

Figure 2 states the decision-layer check in one picture: a discontinuity is evidence for a response switch at a practical boundary, while a smooth curve is evidence that the overt response tracks estimated licensing continuously.

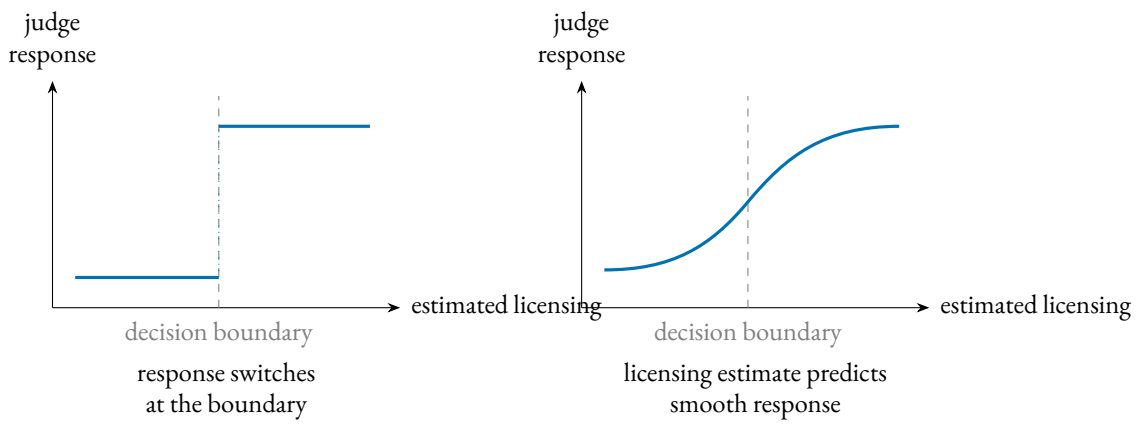


Figure 2: The decision-layer discontinuity check. If asymmetric losses induce a response switch, a judge's treatment of a form should change abruptly as the judge's estimated licensing crosses the practical boundary (left); if the read-out tracks that estimate continuously, treatment should vary smoothly through it (right). A regression-discontinuity design on multinomial hearer responses targets this decision layer, not the constitutive status state itself.

7 CONCLUSION

Grammaticality, de-idealized, isn't one fact about an ideal speaker. It's a conditioned profile: whether some assembly covers a form and its intended value, whether its operator contributions unify, whether its obligatory contrasts are supplied, and whether the relevant population licenses its constructions. A speaker's evidence about that profile and the resulting feelings of anomaly and confidence are separate objects. So are genuine differences among speakers and mere uncertainty about the population mean.

The point of the profile is projection. Weakly evidenced forms should respond differently to framing and exposure from forms repeatedly defeated in genuine opportunities; licensed but processing-heavy forms should improve without a change in licensing; moribund contrasts should lose evidential support before their categorical treatment disappears; and operator errors should draw a different repair profile from value-intact constructional errors. These claims are informative only if situations, niches, alternatives, and target projections are fixed before the outcomes are inspected.

The population dynamics establish the possibility of separated licensing regimes under independently motivated adoption and retention conditions. Evidence updating alone is insufficient. Repair is a candidate maintainer or controller, not one by definition; maintenance requires transmission evidence, and control requires a deviation-sensitive intervention that restores the threatened relation. A history-dependent loss-and-recovery pattern is likewise predicted only where two stable regimes and their external controls have first been established.

The resulting account is less simple than a single grammaticality score, but its recurring distinctions handle cases that otherwise demand unrelated explanations. Their economy has to be earned empirically: estimate licensing from converging non-rating indicators, normalize production by independently specified opportunities, and let the projections fail when their indicators are wrong. De-idealizing grammaticality thereby replaces a tractable fiction with a category that answers to evidence and predicts which rejections soften, which forms transmit, and which absences hold.

A TURKISH VOWEL HARMONY AND OPERATOR-EXPONENT REALIZATION

Turkish illustrates a sharp distinction between LEXICAL disharmony inside stems and ALLOMORPHIC harmony on inflectional suffixes.³ OVMG treats suffixal harmony as operator-exponent licensing and stem disharmony as lexical licensing. Vowel backness isn't a public-update value; plural and past are. In Turkish, the exponents of those operators have to satisfy suffixal harmony.

STEM-INTERNAL VOWELS: DISHARMONY TOLERATED. Loanwords such as *doktor* 'doctor' violate backness harmony, but are fully acceptable; speakers store the form, and the community licenses it consistently.

- (10) *doktor*
 doctor
 'doctor' (disharmonic stem, grammatical)

Because the word contains no malformed operator exponent, a stored construction covers it, the relevant operators unify, and the word is licensed.

Such forms can begin with narrow licensing in borrowing, quotation, or bilingual situations whose participants share access to the foreign expression. Repeated adoption and retention can extend an item beyond that niche until speakers store it as an ordinary lexeme. The result is item-specific licensing of *doktor*. Stem disharmony doesn't become a productive pattern, and the suffixal harmony requirements remain unchanged.

SUFFIXAL HARMONY: OPERATOR-EXPONENT LICENSING. Inflectional morphemes are lexicalized with an underspecified vowel; the licensed allomorph copies stem backness and, for some suffixes, rounding. Using the "wrong" vowel doesn't make the intended value unrecoverable: speakers can identify *-ler* as a plural exponent and *-du* as a past exponent. What fails is licensing of that exponent choice in a dense operator-exponent niche.

- (11) a. *kitap-lar*
 book-PL.BACK
 'books' (harmonic, grammatical)
 b. **kitap-ler*
 book-PL.FRONT
 intended 'books' (suffix harmony violation)
- (12) a. *gül-dü*
 laugh-PST.FRONT.ROUND
 's/he laughed' (harmonic, grammatical)
 b. **gül-du*
 laugh-PST.BACK.ROUND
 intended 's/he laughed' (suffix harmony violation)

OVMG ACCOUNT. For the ill-formed **kitap-ler* and **gül-du*:

³See (Kabak & Vogel, 2001; Sezer, 1981) for discussion of the phonology and (Arik, 2015) for experimental evidence on native judgments.

- The plural and past constructions are securely licensed at the population level. The mismatched exponent choice isn't licensed.
- The mismatched allomorph is still identifiable as an intended exponent, so the token is covered by a candidate operator-exponent assembly. But its exponent-choice node is almost universally excluded by overwhelming preemption from harmonic suffix choices across the many plural and past opportunities. The result is concentrated non-licensing and high confidence, not empty coverage.

Repeated suffix opportunities support the same allomorphic choice, yielding confident categorical rejection. This keeps suffix-harmony errors apart from word salad. **Can the have running* lacks a covering assembly; **kitap-ler* has a recoverable intended value and an identifiable exponent, but the exponent choice is almost universally excluded. By contrast, *doktor* in (10) is covered by its own stored node: stem disharmony violates no operator-exponent licensing condition, so the word remains licensed and the disharmony registers only as a phonological cost in the subjective read-out (§2.6).

LOCALIZED EXCEPTIONS. Some derivational suffixes (e.g. *-imsi* 'ish') are lexically marked "disharmonic". Speakers store them, and community licensing remains consistent: no exponent condition is violated, so the form remains licensed despite vowel mismatch. Clitic boundaries that start a fresh harmony cycle (Kabak & Vogel, 2001) are handled the same way: they satisfy the operator-exponent requirement and leave harmony to phonology alone.

Turkish suffix-harmony errors are grammaticality failures in a precise sense: they don't mis-set a public-update contrast themselves, but they choose an unlicensed exponent for a closed inflectional operator in a dense niche. Stem disharmony lowers phonological well-formedness and affects only the subjective cost vector. A counterexample to the operator-specific response claim would be a matched phonological contrast with no operator-exponent role that produced the same update-confusion and update-oriented repair profile.

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